MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SINGLE HOLE MOUNT, BAYONET COUPLING, SOLDER PIN CONTACT), (SERIES III, CLASSES H AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-83723.

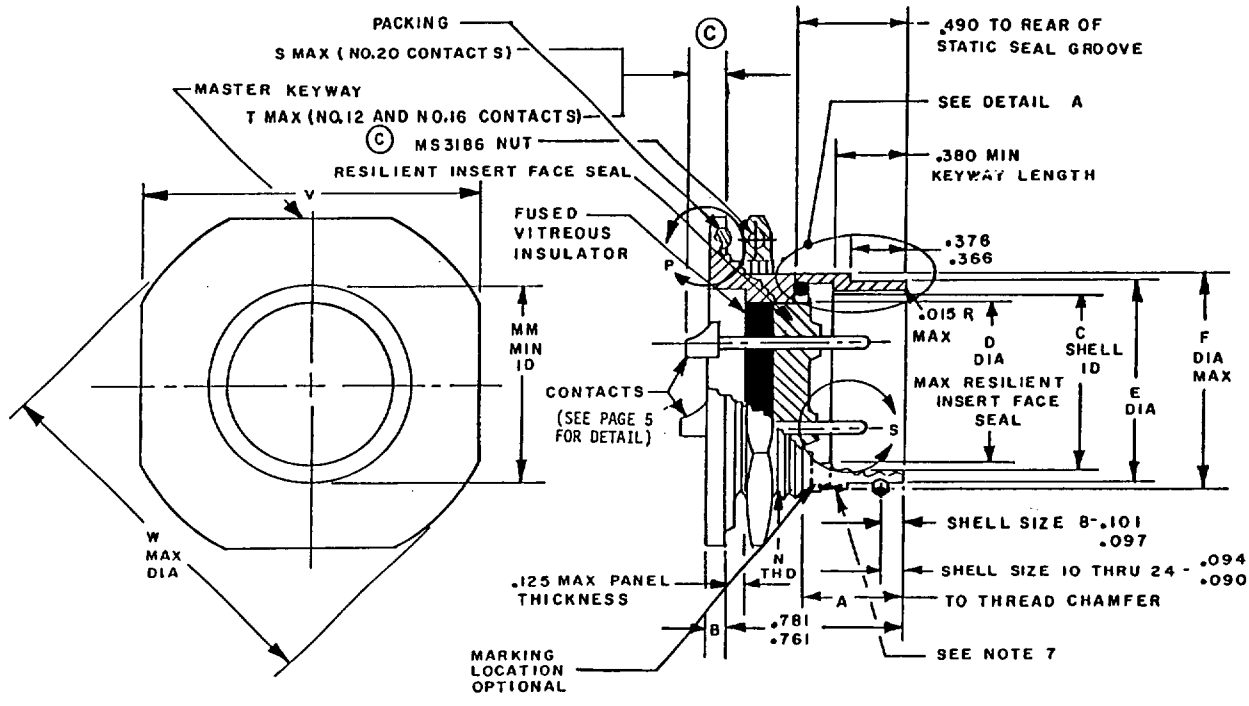


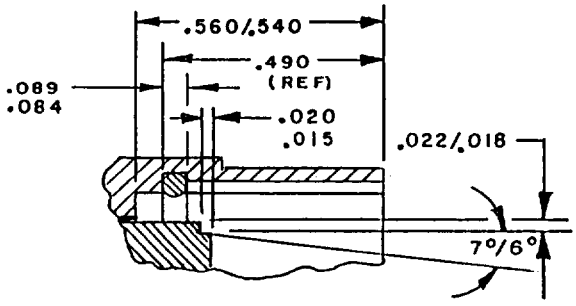
FIGURE 1. Dimensions and configurations.

Ⓒ denotes changes

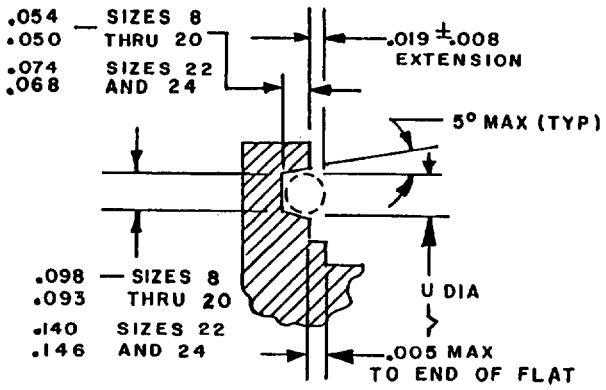
AMSC N/A FSC 5935

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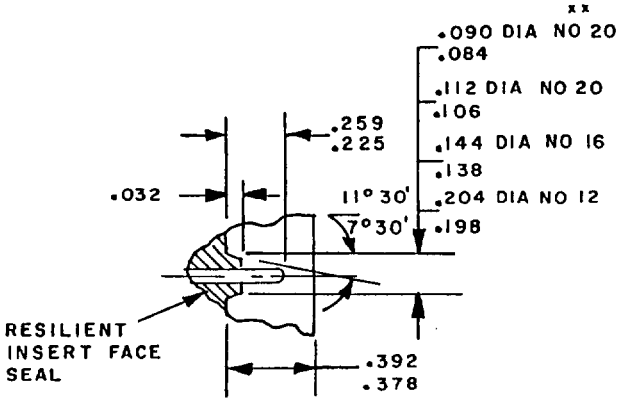
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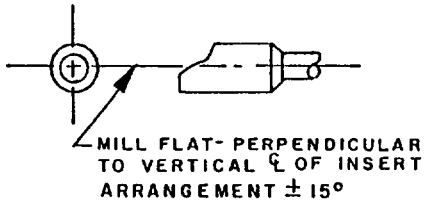
DETAIL A



DETAIL P



DETAIL S



SOLDERPOT ORIENTATION

** SEE TABLE I FOR CONTACT CAVITIES REQUIRING THIS DIMENSION

FIGURE 1. Dimensions and configurations - Continued.

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Connector dimensions

Shell size	A	B	C	D	E	F	N thread	S	T	U	V	W
8	.481 .461	.137 .097	.428 .433	.312	.536 .531	.561	.625-20 UN-2A	.165	.235	.670	.984 .974	1.068
10	.481 .461	.137 .097	.530 .535	.388	.659 .654	.696	.750-20 UNEF-2A	.165	.235	.777	1.109 1.099	1.192
12	.481 .461	.137 .097	.700 .705	.558	.829 .824	.875	.9375-20 UNEF-2A	.165	.235	.965	1.296 1.286	1.380
14	.481 .461	.137 .097	.769 .774	.627	.898 .893	.935	1.000-20 UNEF-2A	.165	.235	1.090	1.396 1.386	1.505
16	.481 .461	.137 .097	.896 .901	.772	1.025 1.020	1.062	1.125-20 UN-2A	.165	.235	1.215	1.521 1.511	1.630
18	.481 .461	.137 .097	1.002 1.007	.860	1.131 1.126	1.187	1.250-18 UNEF-2A	.165	.235	1.340	1.646 1.636	1.756
20	.481 .461	.137 .097	1.127 1.132	.985	1.256 1.251	1.312	1.375-18 UNEF-2A	.165	.235	1.465	1.771 1.761	1.860
22	.481 .461	.168 .128	1.252 1.257	1.110	1.381 1.376	1.437	1.500-18 UNEF-2A	.165	.235	1.574	1.959 1.949	2.068
24	.481 .461	.168 .128	1.377 1.382	1.235	1.506 1.501	1.562	1.625-18 UNEF-2A	.165	.235	1.699	2.084 2.074	2.160

FIGURE 1. Dimensions and configurations - Continued.

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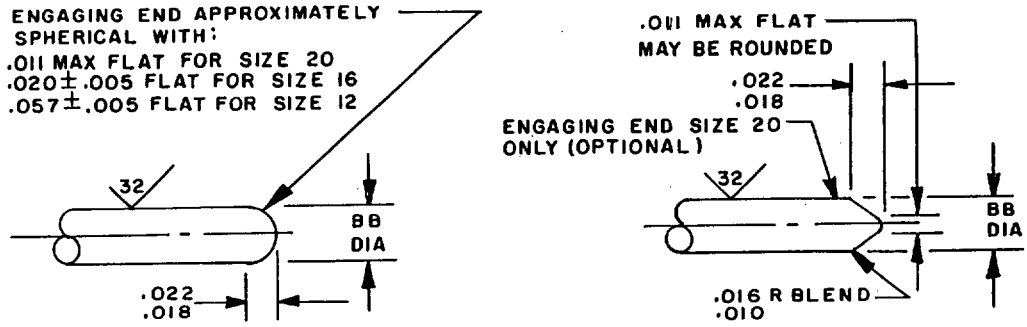
Ⓒ

Shell size	EE	FF	GG	MM	JJ
8	.605	.635	.091 .097	.493	.596 .590
10	.730	.760	.091 .097	.555	.721 .715
12	.917	.947	.091 .097	.743	.908 .902
14	.980	1.010	.091 .097	.805	.971 .965
16	1.105	1.135	.122 .128	.931	1.096 1.090
18	1.229	1.260	.122 .128	1.055	1.220 1.214
20	1.354	1.385	.122 .128	1.243	1.345 1.339
22	1.479	1.510	.122 .128	1.305	1.470 1.464
24	1.604	1.635	.122 .128	1.493	1.595 1.589

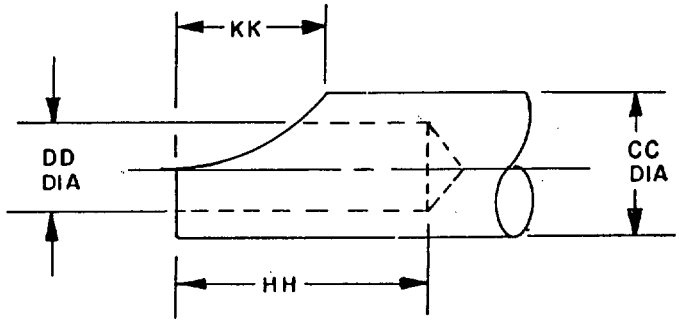
FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:22Z
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ENGAGING END OF CONTACTS



TERMINATION END OF CONTACTS



SOLDERPOT

Dimensions of contact engaging end.

Cont size	BB	CC	DD	HH	KK
20	.041 .039	.088 .061	.048 .042	.188 .109	.110 .068
16	.0635 .0615	.116 .096	.075 .069	.251 .172	.141 .109
12	.095 .093	.151 .131	.122 .112	.251 .172	.141 .109

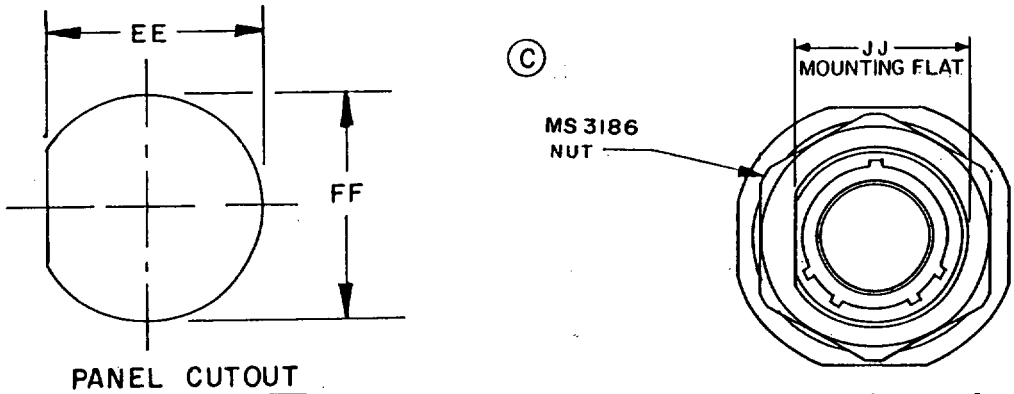


FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:22Z
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Inches	mm	Inches	mm	Inches	mm
.000050	.001	.144	3.66	.721	18.31
.003	.08	.146	3.71	.730	18.54
.004	.10	.151	3.84	.743	18.87
.005	.13	.165	4.19	.746	18.95
.008	.20	.168	4.27	.748	19.00
.010	.25	.172	4.37	.750	19.05
.011	.28	.188	4.78	.756	19.20
.015	.38	.196	4.98	.760	19.30
.016	.41	.204	5.18	.761	19.33
.018	.46	.225	5.72	.769	19.53
.019	.48	.235	5.97	.772	19.61
.020	.51	.251	6.38	.774	19.66
.022	.56	.259	6.58	.777	19.74
.032	.81	.312	7.92	.781	19.84
.039	.99	.366	9.30	.795	20.19
.041	1.04	.376	9.55	.805	20.45
.042	1.07	.378	9.60	.816	20.73
.048	1.22	.380	9.65	.824	20.93
.050	1.27	.388	9.86	.826	20.98
.054	1.37	.392	9.96	.829	21.06
.057	1.45	.428	10.87	.860	21.84
.061	1.55	.433	11.00	.875	22.23
.0615	1.56	.461	11.71	.893	22.68
.062	1.57	.474	12.04	.869	22.76
.0635	1.61	.481	12.22	.898	22.81
.068	1.73	.484	12.29	.900	22.86
.069	1.75	.490	12.45	.901	22.89
.074	1.88	.493	12.52	.902	22.91
.075	1.91	.530	13.46	.908	23.06
.084	2.13	.531	13.49	.917	23.29
.088	2.24	.535	13.59	.918	23.32
.089	2.26	.536	13.61	.920	23.37
.090	2.29	.540	13.72	.931	23.65
.091	2.31	.555	14.10	.935	23.75
.093	2.36	.558	14.17	.9357	23.81
.094	2.39	.560	14.22	.942	23.93
.095	2.41	.561	14.25	.947	24.05
.096	2.44	.576	14.63	.952	24.18
.097	2.46	.586	14.88	.954	24.23
.098	2.49	.590	14.99	.956	24.28
.101	2.57	.596	15.14	.965	24.51
.106	2.69	.602	51.29	.970	24.64
.109	2.77	.605	15.37	.971	24.66
.110	2.79	.620	15.75	.974	24.74
.112	2.84	.625	15.88	.980	24.89
.116	2.95	.627	15.93	.984	24.99
.122	3.10	.635	16.13	.988	25.10
.125	3.18	.654	16.61	.989	25.12
.128	3.25	.659	16.74	1.000	25.40
.131	3.33	.670	17.02	1.002	25.45
.137	3.48	.696	17.68	1.007	25.58
.138	3.51	.700	17.78	1.010	25.65
.140	3.58	.705	17.91	1.020	25.91
.141	3.58	.715	18.16	1.025	26.04

FIGURE 1. Dimensions and configurations - Continued.

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Inches	mm	Inches	mm
1.048	26.62	1.380	35.05
1.055	26.80	1.381	35.08
1.058	26.87	1.382	35.10
1.062	26.97	1.385	35.18
1.068	27.13	1.386	35.20
1.090	27.69	1.394	35.41
1.096	27.84	1.421	36.09
1.099	27.91	1.422	36.12
1.105	28.07	1.432	36.37
1.108	28.14	1.437	36.50
1.109	28.17	1.452	36.88
1.110	28.19	1.455	36.96
1.114	28.30	1.464	37.19
1.125	28.58	1.465	37.21
1.126	28.60	2.470	37.34
1.127	28.63	1.479	37.57
1.131	28.73	1.493	37.92
1.132	28.75	1.500	38.10
1.135	28.82	1.501	38.13
1.142	29.01	1.505	38.23
1.172	29.77	1.506	38.25
1.182	30.02	1.511	38.38
1.187	30.15	1.521	38.63
1.192	30.28	1.545	39.24
1.202	30.53	1.562	39.67
1.214	30.84	1.569	39.85
1.215	30.86	1.570	39.88
1.220	30.99	1.574	39.98
1.229	31.22	1.578	40.08
1.235	31.37	1.579	40.11
1.243	31.57	1.589	40.36
1.250	31.75	1.595	40.51
1.251	31.78	1.596	40.54
1.252	31.80	1.604	40.74
1.256	31.90	1.625	41.28
1.257	31.93	1.630	41.40
1.260	32.00	1.635	41.53
1.286	32.66	1.636	41.55
1.295	32.89	1.646	41.81
1.296	32.92	1.671	42.44
1.298	32.97	1.699	43.15
1.305	33.15	1.705	43.31
1.308	33.22	1.756	44.60
1.312	33.32	1.761	44.73
1.328	33.73	1.771	44.98
1.329	33.76	1.795	45.59
1.339	34.01	1.829	46.46
1.340	34.03	1.860	47.24
1.345	34.16	1.949	49.50
1.346	34.19	1.959	49.76
1.354	34.39	2.068	52.53
1.375	34.93	2.074	52.68
1.376	34.95	2.084	52.93
1.377	34.98	2.160	54.86

NOTES: (Notes 1 through 6 applicable to contacts only).

1. Contact finish: Gold plate .000050 inch mininum. In accordance with MIL-G- 45204. A suitable underplating shall be used. Silver shall not be used. Finish is applied after sealing.
2. Dimensions are in inches. Dimensions apply after plating.
3. 63 all over per ANSI B46. 1-62 unless otherwise specified.
4. Remove all burrs and sharp edges .003 to .005 R.
5. Diameters to be concentric within .004 TIR.
6. Diameters in the same cross-sectional plane to be concentric with each other within .003 TIR.
7. Polarizing stripe shall be color coded blue.
8. Three pins to be located within .004 either side of (TP) relative to master polarizing keyway.
9. Unless otherwise specified, tolerances are .005 (0.13 mm) for three place decimals and .01 (0.25 mm) for two place decimals. Angles ±1°.
10. Metric equivalents are given for general information.

FIGURE 1. Dimensions and configurations - Continued.

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TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

Shell size	Insert arrangements	Contact cavities
8	-3	1, 2, 3
10	-6	1, 2, 3, 4, 5, 6

REQUIREMENTS:

Dimensions and configuration: See figure 1.

This connector mates with MIL-C-83723/75 and MIL-C-83723/77.

For insert arrangements see MIL-STD-1554 and table I.

Part number example:

M83723/81 H 22 55 N

General specification number

Specification sheet number

Class (see MIL-C-83723)

Shell size (see figure 1)

Insert arrangement (see MIL-STD-1554 and table I)

Polarizing position (see MIL-STD-1554)

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, AV, MI
Navy - EC
Air Force - 11, 15, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-3559-03)

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